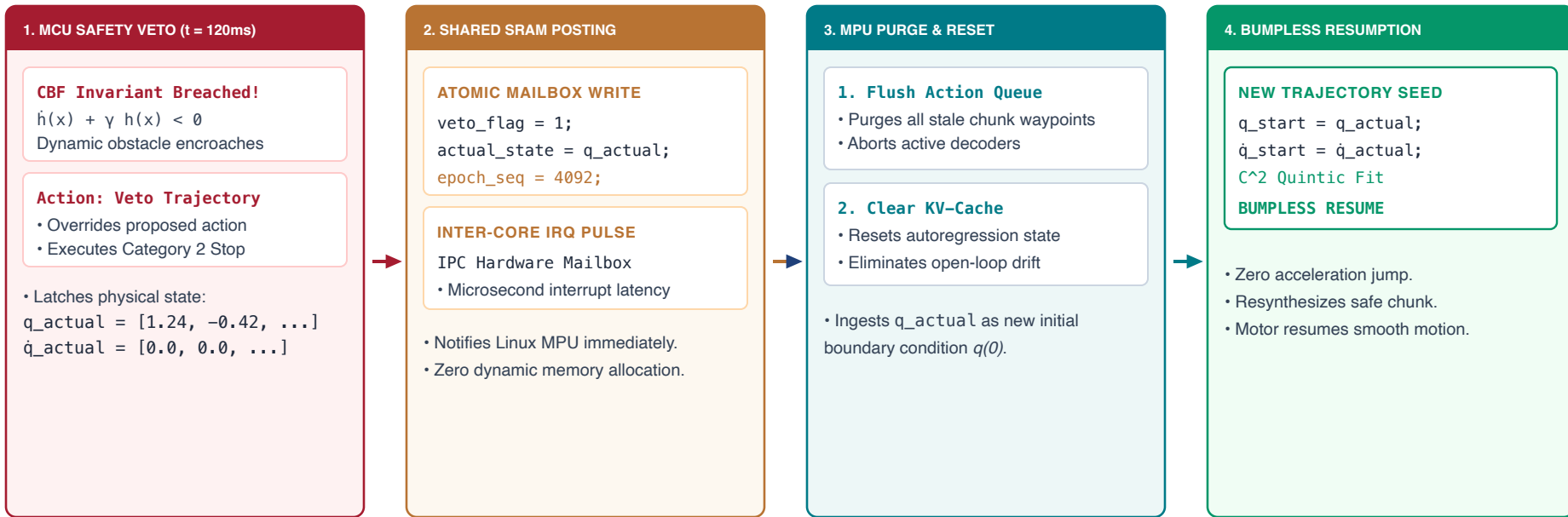


ASYNCHRONOUS STATE RE-ANCHORING HANDSHAKE UNDER SAFETY INTERVENTION

Surviving real-time MCU safety vetoes: atomic mailbox notification, queue purge, KV-cache reset, and bumpless restart



THE PHYSICAL AI STATE RE-ANCHORING CONTRACT

1. **The Hallucination Danger:** If an MPU planner generates trajectories assuming an un-vetoed path, its next chunk commands a massive step jump from the actual physical stopping point.
2. **Strict Hardware Handshake:** By enforcing that every new chunk starts at the exact measured physical state ($q_{\text{actual}}, \dot{q}_{\text{actual}}$), position error is zeroed.
3. **Governed Recovery:** The system gracefully recovers from dynamic obstacles without rebooting the robot or entering an unrecoverable fault state.